

ZAMOKTA, Russian Federation

WMO: 307410

Lat: 52.767N

Lon: 109.967E

Elev: 1286

StdP: 86.80

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -31.2	(c) -29.3	(d) -34.6	(e) 0.2	(f) -30.2	(g) -32.4	(h) 0.2	(i) -28.8	(j) 13.2	(k) -23.7	(l) 11.1	(m) -24.6	(n) 3.8	(o) 300	(p) 0.727

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.8	(c) 25.4	(d) 14.9	(e) 23.6	(f) 14.1	(g) 22.0	(h) 13.8	(i) 17.1	(j) 21.8	(k) 15.9	(l) 20.7	(m) 14.9	(n) 19.4	(o) 3.0	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	12.9	19.0	14.4	12.0	17.6	13.2	11.1	16.6	53.2	21.7	49.4	20.7	46.3	19.4	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	8.6	7.6	-33.4	28.8	3.3	2.1	-35.8	30.3	-37.7	31.5	-39.6	32.6	-42.0	34.1
			-33.1	18.8	3.3	1.2	-35.5	19.6	-37.5	20.3	-39.3	21.0	-41.7	21.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.5	-20.6	-17.6	-10.3	-2.1	5.0	13.1	15.8	13.2	5.8	-3.0	-12.4	-18.0
	DBStd	13.52	4.82	5.77	6.11	5.14	4.90	4.50	3.29	3.69	4.78	5.41	5.38	4.59
	HDD10.0	5003	948	772	630	363	168	21	2	13	143	404	672	868
	HDD18.3	7628	1206	1006	888	612	412	164	92	161	375	662	922	1126
	CDD10.0	439	0	0	0	1	14	113	181	113	18	0	0	0
	CDD18.3	23	0	0	0	0	1	6	13	3	0	0	0	0
	CDH23.3	179	0	0	0	0	6	67	89	16	1	0	0	0
	CDH26.7	23	0	0	0	0	1	11	12	1	0	0	0	0
(14) Wind	WSAvg	3.0	3.1	3.0	3.2	3.5	3.7	3.0	2.4	2.5	2.9	2.9	2.9	2.8
(15) Precipitation	PrecAvg	360	7	5	6	15	24	48	88	74	42	17	18	16
	PrecMax	495	21	11	17	34	59	132	213	150	91	66	28	33
	PrecMin	221	3	2	2	3	7	11	18	25	10	5	7	8
	PrecStd	80	4	2	3	8	13	30	44	32	20	12	6	6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.1	-2.0	6.3	14.8	23.7	28.1	28.4	25.4	21.6	12.4	2.4	-6.4
		MCWB	-9.1	-5.0	0.8	5.2	10.8	15.2	16.5	14.7	12.2	4.9	-0.7	-7.3
	2%	DB	-10.7	-5.5	2.8	10.9	19.3	25.1	25.8	23.1	18.5	9.0	-0.7	-8.9
		MCWB	-11.7	-7.7	-1.7	3.4	9.0	14.3	15.7	14.2	10.7	3.1	-2.8	-9.7
	5%	DB	-12.5	-7.5	0.2	8.3	16.5	23.3	24.0	21.4	15.7	6.5	-2.7	-10.5
		MCWB	-13.3	-9.0	-3.2	2.0	7.5	13.6	14.8	13.7	9.7	1.6	-4.1	-11.1
	10%	DB	-14.1	-9.4	-1.8	5.8	13.8	21.2	22.2	19.6	13.2	4.0	-4.8	-11.9
		MCWB	-14.8	-10.7	-4.7	0.7	6.0	12.8	14.5	13.4	8.4	0.4	-5.8	-12.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.1	-4.9	1.4	5.8	11.8	19.4	18.7	18.2	13.8	5.8	-0.7	-7.2
		MCDB	-8.2	-2.0	5.6	13.8	22.5	23.7	24.4	21.4	19.2	11.6	1.6	-6.7
	2%	WB	-11.6	-7.3	-1.2	3.7	9.5	16.6	17.2	16.2	11.6	3.5	-2.6	-9.6
		MCDB	-10.8	-5.7	2.5	10.1	18.3	21.8	22.3	19.8	16.8	8.2	-1.0	-9.1
	5%	WB	-13.2	-9.0	-3.0	2.4	7.9	14.9	16.2	14.9	10.3	2.0	-4.2	-11.0
		MCDB	-12.5	-7.6	0.1	7.7	15.6	20.3	21.1	18.8	14.5	5.8	-2.7	-10.5
	10%	WB	-14.8	-10.7	-4.6	0.9	6.3	13.6	15.4	13.9	9.1	0.6	-5.8	-12.4
		MCDB	-14.2	-9.4	-1.9	5.6	13.1	19.4	20.0	18.3	12.7	3.7	-4.7	-11.9
(35) Mean Daily Temperature Range	MDBR		6.5	7.8	9.2	9.3	10.2	10.8	9.8	8.9	8.8	7.7	6.6	6.0
	5% DB	MCDBR	8.1	9.3	10.3	11.6	14.5	13.7	12.5	11.7	12.0	10.6	8.8	6.4
		MCWBR	7.3	7.6	7.3	6.8	7.4	6.4	5.3	5.3	6.2	6.2	7.2	5.7
	5% WB	MCDBR	8.0	9.0	10.1	11.0	13.2	11.7	10.0	9.7	10.8	9.9	8.7	6.1
		MCWBR	7.3	7.5	7.4	6.8	7.4	7.3	5.6	5.6	6.4	6.3	7.3	5.6
(40) Clear-Sky Solar Irradiance	taub		0.231	0.255	0.283	0.329	0.396	0.407	0.377	0.343	0.307	0.276	0.252	0.231
	taud		2.219	2.205	2.215	2.268	2.158	2.230	2.373	2.451	2.506	2.497	2.308	2.177
	Ebn at Noon		780	864	908	904	852	845	868	882	882	845	745	697
	Edh at Noon		63	88	109	119	140	132	113	98	82	66	58	54
(44) All-Sky Solar Radiation	RadAvg		1.18	2.20	3.67	5.03	5.71	5.99	5.52	4.67	3.53	2.19	1.20	0.85
	RadStd		0.07	0.09	0.11	0.28	0.33	0.39	0.46	0.25	0.28	0.19	0.09	0.05

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	N/A	-1.73	-2.07	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page